Daniel Panizo | Cover Letter

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Dear employer, Autumn 2023

The purpose of this letter is that of applying for a postdoc position in theoretical physics within your department/division. I am currently a Ph.D. student, under the supervision of Prof. Ulf Danielsson, in the string cosmology group of the department of Physics and Astronomy of the University of Uppsala.

My professional profile is outlined in detail in the enclosed CV and the other files in attachment (available also in my personal webpage). It shows an evergrowing conceptual background knowledge combined with widening communication and computational skills, dressed with enthusiasm and diligence to the research world. Furthermore, it is not only my personal and professional growth, but also those' I surround with, what I seek. This translates to a very positive attitude towards educational and pedagogical activities, such as teaching, supervising students and scientific outreach outside academia.

My research interests include various topics in string phenomenology and cosmology with a particular focus on model building capable of reproducing dark energy from higher dimensions, as well as the relation between swampland conjectures and our current observational evidence. In the immediate future, I would like to continue enhancing the dark bubble cosmology model and start exploring further connections between the swampland framework and cosmological observations. More information can be found in the enclosed file named "Statement of Purpose". Let me now analyse my past career path and all the experiences that took me to this point.

A dagger, designed and sharpen to the point that it cuts the fabric of spacetime, is responsible for my interest in theoretical physics. This object of a fantasy novel (His Dark Materials) is imagined from the idea that our Universe could belong to a set of them, stacked through one extra dimension, only accesible to those carrying such knife. This fictional idea rapidly grew inside me during my high school years, triggering a deep interest in fundamental physics and more specially, in cosmology. Years after, I was firmly entering through the main door of the physics faculty building.

After completing my bachelor in physics at Universidad Complutense de Madrid (Spain), and equipped with a flourishing modest knowledge in fundamental physics, I was accepted to study my master in theoretical physics at Uppsala University (Sweden). A new enviroment, complemented with a master program that provides freedom to learn at each student's pace, boosted further my most fundamental thirst of knowledge. An emerging professional maturity, acquired through academic (and also everyday) experience, was in charge to tame the primeval motivation, decomposing its big questions into smaller, more handeable ones. "Manageable" questions of the form: **Why does the Universe have four dimensions?** or, **Why does the Universe (apparently) expand?** guided my steps towards the string theory framework and its applications to cosmology. As a culmination to the two year quantum fields and strings master program, I devoted 45 out of its 120 ECTS credits to string cosmology aspects:

- My Master thesis, supervised by Prof. Giuseppe Dibitteto, where I explored flux compactifications of ten dimensional supergravities to four dimensional effective Anti de Sitter (AdS) vacua. Although the results could mathematically differ in aspect from each other, they can secretly represent the same physics. If they do, there is always a set of reparametrisations (dualities) that can be performed to connect those solutions.

- Furthermore, I decided to extend my studies in dualities with an addendum-like 15 ECTS project, where I studied the (non) geometrical and algebraic properties of generalised fluxes both in IIA and IIB supergravity theories. This project was supervised by Prof. Dieter Lüst, who personally acepted my

research proposal, and thanks to an Erasmus+ program grant, I became a visitor student at Ludwig Maximillian University in Munich during autumn 2018. In addition to the European grant, I was awarded with the Swedish Lundström-Åman grant, which allowed me to extend the project and research during spring 2019.

These two projects, although exciting and thirst-knowledge allieviating, were not enough. As it is the norm in research, the deeper you look down into a question, the more it scatters into more fundamental ones. It requires to go beyond in erudition, which continues by doctoral studies.

It was spring 2019, when I was informed that the Center of Interdisciplinary Mathematics (CIM), which belongs to Uppsala University's Mathematics department, was willing to allow me present my own Ph.D. research proposal to its committee. This was an unique procedure, as it is normally the potential supervisor (and not the candidate) who proposes the research plan and the CIM board decides if it is worth to offer to applicants. I think the nominating board liked my proactivity and research plan proposition, which in addition to the support of my current supervisors (Prof. Ulf Danielsson, Prof. Tobias Ekholm and Prof. Georgios Dimitroglou), resulted in my proposal being elected to become a Ph.D. position. This I happily accepted, as I had already fallen in love with the peaceful life style Uppsala provides, after two years and a half living in the city. A feeling that gently pushed me to learn the beautiful language that Swedish is, which helped me to become fully integrated in the local enviroment.

I have been extremely lucky to continue researching along my primal adolescent motivation during these last four years. Together with Prof. Ulf Danielsson, and with esporadic collaborations with Prof. Thomas Van Riet and other researchers across Europe and USA, we have been able to push the boundaries of the string cosmology state of art, by enhancing an alternative to compactifications, in the form of a braneworld scenario called **dark bubble cosmology** (more details in the statemente of purpose). We have made the model grow robust, with each of its features compatible with observations. Without hesitation, I would affirm that my greatest achievement is that of modeling a consistent "top-down" construction of dark bubbles from string theory. Every time the puzzle pieces of each research project start to fit, joy and shivers rank up my passion for string phenomenology.

Over my Ph.D. years, simultaneously to all research, I decided to actively broaden my knowledge across disciplines beyond physics, and I took courses in philosofy, religion, mathematics, programming, economics, Swedish... which have granted me new technical capabilities and different point of views in respect to research implications and academic work. Furthermore, I have enthusiastically performed my duties as teacher assistant of subjects like electrodynamics and general relativity, selecting the most pedagogical examples and applying my best didactical competences when teaching or tutoring to the students, aiming to provide them the best possible learning.

And this is where I stand now. In the meantime my research and personal interests got broader and broader and my research line is maturing. I feel ready to take the next step in my research career and become a postdoc, with all the associated responsibilities and exiciting new research to unriddle. I do think that we are living in an exciting era for our understanding of the cosmos, which sooner than later will be boosted further if the scientific community, which I would like to continue being part of, carries on unraveling the Universe and its mysteries.

Yours faithfully,

Daniel Panizo

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Education

2019–2024 **Doctor of Philosophy**, *Uppsala University*, Uppsala, (Sweden)
thesis title *(Preliminary) Entre dos vacua: Bubbles and Bulk*
supervisors Ulf Danielsson, Tobias Ekholm, Georgios Dimitroglou
description Embedding of the dark bubble model in string theory.

2016–2019 **Degree of Master of Science (120 ECTS)**, *Uppsala University*, Uppsala, (Sweden)
thesis title *Type IIB compactifications and string dualities*
supervisors Giuseppe Dibitetto
description Compactifications on toroidal orbifolds, non-geometric fluxes and S,T dualities

2010–2016 **Bachelor in Physics (240 ECTS)**, *Universidad Complutense de Madrid*, Madrid, (Spain)

Publications

[1] U. Danielsson and D. Panizo, *Experimental tests of dark bubble cosmology*, 2311.14589.

[2] I. Basile, U. Danielsson, S. Giri and D. Panizo, *Shedding light on dark bubble cosmology*, 2310.15032.

[3] U. Danielsson, O. Henriksson and D. Panizo, *Stringy realization of a small and positive cosmological constant in dark bubble cosmology*, *Phys. Rev. D* **107** (2023) 026020 [2211.10191].

[4] U. Danielsson, D. Panizo and R. Tielemans, *Gravitational waves in dark bubble cosmology*, *Phys. Rev. D* **106** (2022) 024002 [2202.00545].

[5] U.H. Danielsson, D. Panizo, R. Tielemans and T. Van Riet, *Higher-dimensional view on quantum cosmology*, *Phys. Rev. D* **104** (2021) 086015 [2105.03253].

Awards and Scholarships

2019–2024 **CIM Scholarship**, *Uppsala University*
PhD funding awarded by CIM to attend conferences, visit other institutions, as well as acquiring a computer.

2019 **Lundström-Åman scholarship**, *Uppsala University*
Project: Compactifications on Calabi-Yau manifolds: A backstage for string cosmology
Master funding awarded byUppsala University to pay for accomodation expenses while the associated project was conducted.

2018 **Erasmus+ scholarship**, *European Commission*
Erasmus funding awarded by Erasmus+ programme to pay for accomodation expenses during my research visit to Arnold Sommerfeld Center (Munich) under Dieter Lüst supervision.

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Talks

- (Invited) *Experimental tests of dark bubble cosmology*, (digital) seminar at IFT, Madrid (Nov 2023)
- (Invited) *Our Universe on a (Dark) Bubble*, (digital) seminar at CAPT, University of Nottingham (Nov 2023)
- (Invited) *A cosmological excursion to Dark Bubbles*, (digital) talk at String Pheno seminars (Nov 2023)
- (Invited) *Our Universe on a (Dark) Bubble*, (digital) seminar at Oviedo University Physics Department (Nov 2023)
- (Invited) *Dark bubble cosmology*, (digital) seminar at Kyoto University (Jul 2023)
- (Invited) *Blowing bubbles from string theory*, seminar at Helsinki Institute of Physics (Dec 2022)
- (Contributed) *Dark bubbles and their string embedding*, talk at the 31st Nordic Network meeting on "Strings, Fields and Branes" in Copenhague (Nov 2022)
- (Contributed) *Riding bubbles to take off from the swamp*, talk at String phenomenology conference (Jul 2022)

Teaching experience

2020–2024	**Teacher Assistant**, *Classical Electrodynamics (10 ECTS)*
2022–2023	**Teacher Assistant**, *Gravitation and Cosmology (10 ECTS)*
2022–2023	**Lab Assistant**, *Mecanic III (10 ECTS)*

Computer skills

Mathematica	■■■■□
Latex	■■■■□
Python	■■■□□
html	■□□□□

Languages

Spanish	Native
English	C1
Swedish	B2
French	A2
Other language	Willing to learn

Referees

- Ulf Danielsson, Professor at Uppsala University (Sweden)
- Thomas Van Riet, Associate Professor at KU Leuven (Belgium)
- Pietro Longhi, Senior Lecturer at Uppsala University (Sweden)

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Daniel Panizo | Statement of Purpose

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Recent research activity: Blowing bubbles to escape from the swamp

The swamp is a cold place, fill with mud, mosquitos and dangers everywhere you step on. Unfortunately, it seems that our understanding of cosmology from a string theory perspective lies in a sludgy puddle of these lands. Is there any way we can make that knowledge escape to better grounds? Unraveling around this question has been my main interest during my Ph.D. studies.

Although the last decade has resulted in great progress towards the construction of realistic models of cosmology from string theory, there are some major issues that remain. To build models where all moduli are stabilised, equipped with scale separation and a small and possitive cosmological constant (CC) is still an unsolved problem [1]. Concurrently, based on the evidence pointing to the fact that less than expected low dimensional effective field theories (EFT) could be obtained from quantum gravity, the **swampland** program was created [2]. One of the most striking conclusions that may be drawn from this criteria is that de Sitter (dS) cosmology may belong to the swampland.

At this point, one question has to be raised: Is there any way to scape from the swampland? An alternative mechanism to **flux compactification** is **braneworld** cosmology. My Ph.D. research has been devoted to develop a braneworld model, which makes use of swampland constrains, i.e. the Weak Gravity Conjecture (WGC), as its foundation. This model is known as **dark bubble** cosmology [3].

In simple grounds, this model is the realisation of an **induced** four dimensional dS cosmology living on a three spatial dimension brane. This mediates the decay of a five dimensional true vacuum that has **nucleated** inside a false one. The region of vacuum emcompased by the brane is colloquially called **bubble**, which will nucleate at rest and finite size, as a **Brown-Teitelboim** instanton. If the bubble nucleates with a subcritical tension, it will then start to expand. The value disagreement between the tensions of the bounce solution (i.e. critical) and the nucleated bubble can be interpreted as a positive CC on the induced cosmology, giving name to the model; Four dimensional **dark energy** is no more than a dynamical property of a higher dimensional expanding bubble. Together with my supervisor, Ulf Danielsson, Thomas van Riet and other collaborators, I have also explored how the brane **backreacts** on the bulk geometry, which in turn affects the induced effective gravity and energy-momentum tensor on the first, resulting in familiar energy densities decorating the induced cosmology, such as gravitational and electromagnetic radiation [4, 5]. **Gauss-Codazzi** equations are the dictionary to understand how gravity, extending along the bulk, dictates how the aforementioned and brane relate one another.

The derivation of a complete **top-down** construction of the dark bubble model from string theory has been the greatest achievement of my Ph.D. scientific production [6]. In this set-up, the ten dimensional background consists of a stack of D3 branes with angular momentum and **near horizon** limit description as $AdS_5 \times S^5$. **Supersymmetry** is broken due to non-zero temperature and a large chemical potential in the $\mathcal{N} = 4$ Yang-Mills dual description, pointing to an instability that will eventually cause the nucleation of a bubble (which can be identified with the creation event in Vilenkin **quantum cosmology** [7]). This corresponds to the creation of an induced four dimensional cosmology as the brane emerges after tunneling through the potential barrier and begins to expand. Corrections to the DBI action governing the motion of the expanding brane will yield an effective positive, yet small CC in the induced cosmology. This whole construction comes also equipped with a novel energy scale hierarchy [8], where **scale separation** is implemented via tension-to-charge ratio of the expanding brane, rather than small extra dimensions.

Future research

In the near future, I would like to continue researching on phenomenological implications on low dimensional EFT's that can be obtained from string theory. There are two main research territories that I would like to immediately adventure in after obtaining my Ph.D.

The dark bubble formalism is a compelling novel alternative framework that can evade the most prominent issues discussed above when constructing realistic models of cosmology from string theory. This model provides a new paradigm for string phenomenology, with plenty of puzzles still to delve into. For example, one of the most important tasks within dark bubble cosmology would be to understand the uplift of the **standard model** of particle physics into the higher dimensions. This is a highly non-trivial task, which would involve a rigorous study on the dynamics of branes in the compact dimensions of S^5 and strings stretching between them, giving rise to massive sectors in the expanding cosmology. Another bubble aspect that I would like to study is the embedding of the dark bubble in different **supergravity** backgrounds. This analysis would provide further insights about the key features of the model and enhance our understanding towards a more general top-down construction for it.

It is remarkable that, althought the swampland program started as a formal identification of universality properties of consistent quantum gravitational theories, it has spread towards phenomenological aspects in contact with observations over the years. These consistency requirements are challenging our **Naturalness** concept, pointing to the need of constructing EFT's that are sensitive to changes in its ultraviolet (UV) completion. An area that I would like to potentially investigate is how the Swampland Distance Conjecture (SDC) and the WGC affect the inflation mechanism, imposing severe constraints in its derivation from quantum gravity. These restrictions, together with enhanced observational data, may provide insights about the unknown early universe. It is also worth considering the interplay between dark energy, dark matter and the extra mesoscopic **dark dimension** proposed in [9]. This new scenario can shed some light on our comprehension of the dark sector, given the suggested relation between fundamental interactions in our brane and its embedding in bulk space.

References

[1] U.H. Danielsson and T. Van Riet, *What if string theory has no de Sitter vacua?*, *Int. J. Mod. Phys. D* **27** (2018) 1830007 [1804.01120].

[2] C. Vafa, *The String landscape and the swampland*, hep-th/0509212.

[3] S. Banerjee, U. Danielsson, G. Dibitetto, S. Giri and M. Schillo, *Emergent de Sitter Cosmology from Decaying Anti–de Sitter Space*, *Phys. Rev. Lett.* **121** (2018) 261301 [1807.01570].

[4] U. Danielsson, D. Panizo and R. Tielemans, *Gravitational waves in dark bubble cosmology*, *Phys. Rev. D* **106** (2022) 024002 [2202.00545].

[5] I. Basile, U. Danielsson, S. Giri and D. Panizo, *Shedding light on dark bubble cosmology*, 2310.15032.

[6] U. Danielsson, O. Henriksson and D. Panizo, *Stringy realization of a small and positive cosmological constant in dark bubble cosmology*, *Phys. Rev. D* **107** (2023) 026020 [2211.10191].

[7] U.H. Danielsson, D. Panizo, R. Tielemans and T. Van Riet, *Higher-dimensional view on quantum cosmology*, *Phys. Rev. D* **104** (2021) 086015 [2105.03253].

[8] U. Danielsson and D. Panizo, *Experimental tests of dark bubble cosmology*, 2311.14589.

[9] M. Montero, C. Vafa and I. Valenzuela, *The dark dimension and the swampland*, *Journal of High Energy Physics* **2023** (2023) 1.